



Siemens home automation

know that the ambient electronics era has begun!

Promising research, towards an intelligent future

There is lot of interesting research happening in the area of ambient intelligence towards improving the base technologies as well as developing products. The Massachusetts Institute of Technology (MIT), Stanford University, Carnegie-Mellon University, IMEC Belgium, and our own Indian Institutes of Technology (IITs) are world-renowned for their work in this space. Companies such as Philips and Xerox also continue to work with these universities to take the ambient intelligence dream forth.

Fluid interfaces to help you live easier. MIT's ambient intelligence research group (called the Fluid Interfaces Group), for instance, focuses on the human-machine interactive experience—a key aspect of ambient intelligence.

They question "Why do we still use a keyboard and mouse to interact with digital information?" These traditional input-output modes constrain our ability to access and interact naturally with digital content. Traditional computing

systems lack the contextual knowledge to offer relevant information when and where we need it. Further, traditional screen-based interfaces divert our attention from other tasks, especially when we are driving or talking to others. More so, they are designed for single-user applications, like laptops and mobiles, and are not the perfect choice for collaborative activities or public usage.

"These are the problems that motivate our research. Our group designs new interfaces that integrate digital content in people's lives in more fluid and seamless ways. Our aim is to make it easier and more intuitive to benefit from the wealth of useful digital information and services," writes their principal investigator Pattie Maes.

The group works towards augmenting a person's surroundings with relevant digital information, in as inconspicuous a way as possible. Users can not only view but also interact with that information, in a very natural way, without requiring any conscious effort. They also try to alter everyday objects by embedding sensors, actuators and displays so that the objects can respond to users in meaningful, noticeable ways. The most exciting is their invention of interfaces and machines for control and

manipulation of materials such as paper, fabric, wood and food. As the team puts it, their goal is to endow materials and manufacturing with some of the advantages associated with the digital world, such as modifiability, programmability, responsiveness and personalisation. In short, they are trying to merge the best of both worlds!

They have invented a lot of things: small objects with common sense that can play with children and help them learn; mobile applications that watch people's behaviour and give them a timely nudge (what to say and what not to say, for instance) or reminder, as well as help them rethink their actions; tools that help rebuild communities after crises by collecting the residents' talents; tool for spatially organising mails, tasks and mind-maps and comparing these with others'; platform to support full-body interaction with digital content; soft and permeable shutters for environmental control and communications; wearable gesture-based interface for natural interaction with the digital world; invisible mouse; tangible way of composing music using everyday objects; airborne reporting drone; tool for visualising content themes in streaming data; device that fuses real and electronic books; paired objects designed to respond to synchronous movement; touch-based personalisation for ubiquitous computing; and more.

Most of these devices do not look like devices at all. They float in the air, display on the wall and use real-time objects. So you would not feel intimidated by the idea of having so many devices around you.

Understanding your needs better. Stanford University's Ambient Intelligence Research (AIR) Lab is designed in the form of a smart room. While it looks like a very comfortable dwelling, it also offers an ambient interface to its user through pervasive sensing, processing, and communication. The various gadgets in the room are context-driven, and can be personalised to a great extent. They try to understand the user's routine and preferences by watching the